

Darren Zheng

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EDUCATION

University of Pennsylvania, College of Arts and Sciences

Bachelor of Arts in Mathematics, Minor in Statistics, Ben Franklin Scholar

Aug 2021 — May 2025

B.A. GPA: 3.51/4.0

PROJECTS

Independent Study

Sept 2025 — Dec 2025

- Reading "Fourier Analysis & Hausdorff Dimension" by Pertti Mattila under Prof. Phillip Gressman
- Short write-up regarding the proof of the Wolff-Erdogan Theorem can be found on my personal website [here](#)

Undergraduate Honor's Thesis | [Link to PDF](#)

June 2024 — May 2025

- Constructed intrinsic Morse functions on Grassmannian manifolds to compute its Morse homology under Prof. Dennis DeTurck

Directed Reading Program | [Link to Project Write-ups](#)

Aug 2022 — May 2025

- Semester-long independent study with a graduate student. Prepared a presentation/write-up at the end of each semester to rest of participants and mentors. Participated over 6 semesters
 - * Knot Invariants using Algebraic Topology (Yi Wang)
 - * Understanding Generating Functions (Xinxuan Zhang)
 - * Pontryagin Duality of LCA Groups (Travis Leadbetter)
 - * The Distinct Distances Problem (Jae Ho Choi)
 - * Besicovitch Sets & Hausdorff Dimension (Hunter Stufflebeam)
 - * Studying Mapping Class Groups (Ellis Buckminster)

EWS Detection using Topological Data Analysis | [Link to PDF](#)

Jan 2024 — Apr 2024

- Applied and implemented a persistent-homology-based methodology from a paper by Gidea & Katz in Python & R to study the time-series of certain stock market indicators as potential early warning signs of market collapse

EXPERIENCES

Penn Math Contest Club | *Co-founder & President*

Sept 2021 — May 2025

- Co-founded club during freshmen year, focused on hosting a math contest for high schoolers
- Wrote original problems for the test, organized day-of logistics and events, conducted outreach with local Philadelphia schools, communicated with sponsors to secure funding, and wrote handbooks for future organizers
- Spring 2025 contest had ~150 in-person and ~200 online participants; [More details on website here](#)

Penn Undergraduate Math Society | *President*

Jan 2021 — May 2025

- Joined board as freshmen liaison, president during senior year. Helped run events for all math undergraduate majors and broader community
- Organized events include: Course & Major Advising, Graduate Student talks, Integration Bee, Department Merch Sale, Origami Night, Weekly Cookie Office Hours, DRP x PUMS mini-talks, faculty lunches, End-of-Semester Socials

Putnam Club | *Organizer*

Aug 2021 — Dec 2024

- Organized club that meets weekly on Sundays in the fall to practice for the Putnam Exam

WORK

Ross Mathematics Program | *Administrator*

June 2024 — July 2024

- Was an administrator during the 6-week math camp for high schoolers
- Handled various organizational tasks, coordinating social events for the students, and preparing various mathematical talks, including a week-long introduction to point set topology talk series for the students

AwesomeMath Summer Program | *Teaching Assistant*

June 2022 / June 2023 / June 2025

- Was a TA for three summers. Did 2 courses per each 3-week session for 9 weeks each summer
- Hosted problem sessions, office hours and graded daily HW for classes between 10-30 students
- Taught primarily combinatorics I/II, algebra III/IV, and number theory III/IV. See specific course details [here](#)

TEACHING

University of Pennsylvania Math Department | *Teaching Assistant*

Aug 2022 — May 2025

- Responsible for grading assignments/exams, weekly office hours and recitations for most classes
 - * Grader for MATH3200 - Linear Algebra (Jan 2025 — May 2025)
 - * TA for MATH1410 - Calculus II (Sept 2024 — Dec 2024)
 - * TA for MATH2600 - Honors Calculus II (Jan 2024 — May 2024)
 - * TA for MATH1610 - Honors Calculus I (Sept 2023 — Dec 2023)
 - * TA for MATH1410 - Calculus II (Sept 2022 — Dec 2022)

VOLUNTEERING

Science Olympiad at Penn | *Test Writer*

Aug 2021 — May 2025

- Wrote tests for It's About Time, Fermi Problems, Optics, and Codebusters categories for high school competitors over 4 years
- Joined finance board during 2023 — 2024 to secure sponsorship funding for the high-school contest

Matriculate | *Mentor*

Aug 2022 — May 2024

- Mentored first-generation and low-income high school seniors through the year-long college admissions process
- Scheduled weekly meetings and oversaw 3 cohorts (12 students in total)

Princeton Prison Teaching Initiative | *Grader*

Jan 2024 — May 2024

- Graded HW + Exams for a statistics class

Bridge to Math | *Volunteer*

Aug 2022 — May 2025

- Hosted some additional quiz/finals prep sessions throughout the semester as organized by Bridge-to-Math club @ Penn

AWARDS

Good Teaching Award for MATH2600

May 2024

Good Teaching Award for MATH1610

Dec 2023